

**Articulated Concrete Block Data Report
2004 Testing for International Erosion
Control Systems, Inc.**

Prepared for:
IECS, Inc.

Prepared by:
Michael D. Robeson, Christopher I. Thornton, Chris E. Clopper, David Varyu, Victor Barchers,
and Ian Maycumber

January 2005

Colorado State University
Engineering Research Center
Fort Collins, Colorado



2 TEST PROGRAM

2.1 TEST FACILITY

Laboratory testing was performed at the Hydraulics Laboratory of Colorado State University, located at the Engineering Research Center (ERC). Colorado State University's ERC is comprised of laboratories and offices encompassing virtually all engineering disciplines. Within the ERC, the Hydraulics Division, a subdivision of the Civil Engineering Department, operates the Hydraulics Laboratory, a Hydromachinery Laboratory, and a variety of outdoor facilities.

Water supply to both outdoor and indoor research facilities is furnished by Horsetooth Reservoir adjacent to the ERC, as shown in Figure 2.1. Flow is conveyed through an existing pipe network to each facility. Outdoor facilities are gravity fed from Horsetooth Reservoir with a capacity of approximately 170,000 acre-feet of water and provides a maximum, static pressure of approximately 110 pounds per square inch at the ERC pipe network. Each outdoor facility has its own independent water delivery system.

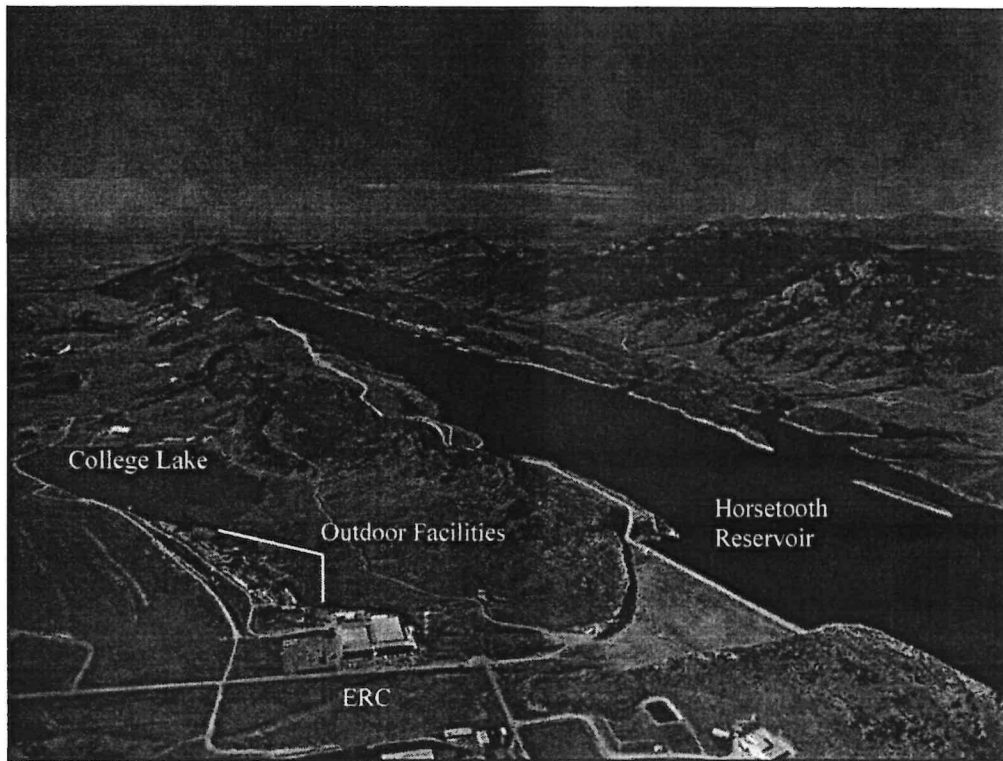


Figure 2.1. CSU Engineering Research Center and Horsetooth Reservoir

For this testing program, a steep-gradient overtopping test facility was utilized. The facility utilized was 4-feet wide by 100-feet long, set at a 2H:1V slope. Headbox configuration in this flume allowed for a maximum of 5 feet of overtopping depth. Figure 2.2 presents a profile drawing of the facility utilized for the testing program.

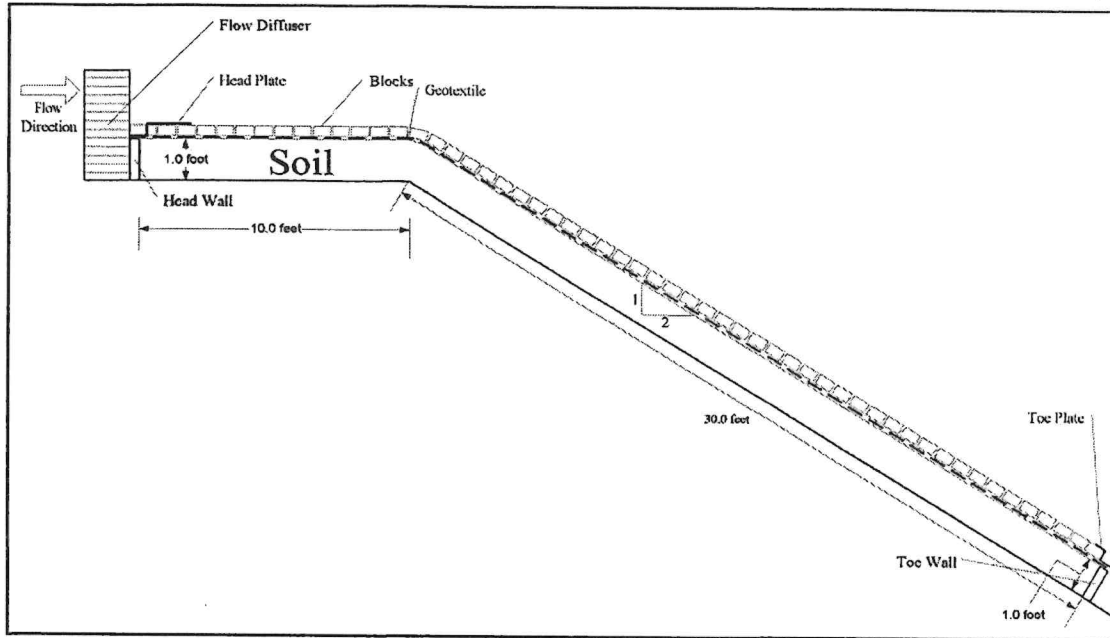


Figure 2.2. Profile View of the Steep-Gradient Overtopping Flume Set at 2H:1V

2.2 PRODUCTS

Two ACB systems from IECS, Inc. were tested in an effort to provide data consistent with the current state-of-the-practice in calculating hydraulic performance thresholds. For clarity of presentation, the physical characteristics and a photograph of each of the ACB systems tested during the program are presented in Table 2.1.

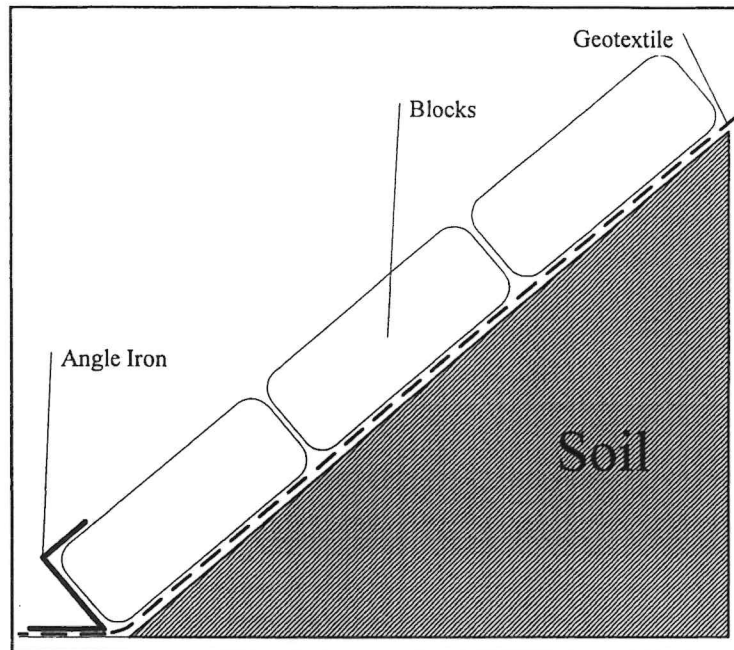


Figure 2.5. Profile View of Toe Detail for ACB Testing

2.5 TEST PROCEDURE

A test consisted of a continuous four-hour flow over the revetment system at a uniform discharge. The performance threshold for the ACB system tested within this test program was defined as the point in which deformation, soil loss, or loss of intimate contact with the embankment subgrade occurred. Provided that the revetment system successfully survived the four-hour flow without exceeding the defined performance threshold, the procedure was repeated at the next higher target discharge or until the flow capacity of the testing facility was reached. Typically, target discharges corresponded to predetermined overtopping depths above the embankment crest elevation (e.g., 1-foot, 2-foot, etc.), although any discharge could be conveyed through the system provided proper measurement and reporting procedures were followed. If

minor system deformation occurred during the test, hourly data collection was maintained for the entire four-hour test duration, unless the performance threshold was exceeded.

Hourly measurements of water surface elevation were made at approximate 2-foot intervals (stations) along the centerline of the embankment during each test. Bed elevations (top of revetment surface) were established prior to and again after the cessation of each test, at the same measurement stations as the water surface readings. Blocks were identified with a mark to ensure consistency in measurement. Flow and elevation measurements were made to the nearest one hundredth (0.01) of a foot using a point gauge and survey level. Stationing positions were established so that the horizontal and lateral location of each measurement station did not vary between subsequent measurements.

Hourly measurements of point velocity were made at two-tenths, six-tenths, and eight-tenths depth of flow, measured down from the water surface, at 2-foot intervals along the centerline stationing, and corresponded to each water surface measurement station. In areas where depth was insufficient (i.e., less than 0.5 ft) to provide velocity measurements at three depths, one measurement at six-tenths depth was made. Velocity measurements were made with a one-dimensional electromagnetic Marsh-McBirneyTM current meter accurate to $\pm 3\%$. The operational scale of the current meter ranges from 0.5 ft/s to 20 ft/s during uniform flow conditions. The axis of the device was maintained at an angle normal to the plane of the embankment while the measurement was made.

The total discharge was determined independently of the measurements made within the test section. Determination of the total discharge was made by using an in-line sonic flow meter placed in the inflow pipe accurate to $\pm 3\%$.

1. Introduction

International Erosion Control Systems (IECS) has been in business since 1984 and has successfully installed more than 16 million square feet (1.48 million square meters) of revetment blocks. The Company has worked with the University of Windsor, the University of Minnesota and Colorado State University over the past 20 years to conduct a variety of hydraulic tests on both full scale and model revetment blocks. Both overtopping flow testing as well as wave impact testing has been conducted. IECS is not aware of any other revetment block manufacturer who has conducted wave impact testing. This report summarizes the various tests conducted and the development of design summary tables and numerical models to predict the Factor of Safety for a given block under specified hydraulic conditions.

Overtopping tests were conducted in accordance with the procedures outlined in “Minimizing Embankment Damage During Overtopping Flow”, Federal Highway Administration Report No. FHWA-RD-88-181 (November, 1988) and “Hydraulic Stability of Articulated Concrete Block Revetment Systems During Overtopping Flow”, Federal Highway Administration Report No. FHWA-RD-89-199 (November 1989).

Wave impact testing and measurement was conducted in accordance with “Coastal Engineering Manual”, U.S. Army Corps of Engineers Manual EM 1110-1100 (as amended up to August, 2008). In addition, block wave impact testing was also compared to analytical results generated by the Anamos Stability of Block Revetment program developed by Delft Hydraulics (The Netherlands).

2. University of Windsor Testing – 1988

This series of tests was conducted on 1:10 scale model blocks of the CC70 revetment block when subjected to overtopping flows. Tests were conducted in a flume measuring 40 feet long, 24 inches deep and 18 inches wide with blocks resting on a horizontal level bed.

The blocks were tested with both the broad face up and the broad face down to determine which orientation provided greater stability of the blocks.

Subcritical and supercritical flows were tested and with and without geotextile filter cloth attached to base of the blocks.

The testing clearly demonstrated that the blocks were more stable with the broad face resting on the channel bed. For all overtopping tests, success was achieved by ensuring that the block maintained contact with the bed at all flows.

Test data also resulted in information related to Manning's n (roughness coefficient) as the depth of the overtopping changed.

Wave impact testing was conducted in the 50 foot long, 2.0 foot wide, 3.6 foot deep wave flume at the University of Windsor Hydraulics Laboratory. For the wave impact tests, 1:4 scale model revetment blocks of the CC20 and CC70 blocks were conducted. Bed slopes of 1.5H:1V, 2.0H:1V and 3.0H:1V were tested. As in the overtopping tests, testing with and without the geotextile filter cloth was conducted. Wave heights ranged from 0.67 to 3.0 feet (prototype) and wave periods ranged from 1.3 to 1.7 seconds.

The wave impact testing resulted in developing a series of curves relating to wave steepness and uplift coefficients which are functions of the wave height, wave length, block height and revetment angle.

3. University of Minnesota Testing – 1999

Testing of full size CC35 and CC45 oversized revetment blocks was conducted at the University of Minnesota St. Anthony Falls Laboratory. The test flume measured 253 feet long, 9 feet wide and 6 feet deep.

In this series of tests, geotextile filter cloth was attached to the base of the blocks for all tests. Overtopping flow depths ranged from 1.0 to 3.0 feet. Bed slopes tested included a horizontal level bed and 2.0H:1V.

The blocks were tested to shear stresses up to 38.5 lbs./sq.ft. and velocities up to 19.0 ft./sec. The test blocks remained stable under all of the test conditions.

4. University of Windsor Testing – 2004

In this series of tests, 1:6.5 scale model revetment blocks were tested for wave impact performance. CC65 open cell, CC95 closed cell and CC90 closed cell oversized blocks were evaluated at the University of Windsor Hydraulics laboratory wave flume facility.

For this series of tests, blocks were evaluated with and without geotextile filter cloth attached to the base of the blocks. Impact wave heights ranged from 3.7 feet to 5.2 feet and wave periods ranged from 3.2 to 5.7 seconds. The blocks were